

Sky Knowledge (20 points)

The projector will display the sky as seen from near the equator (0°N, 19°E). The rotation of the sky will be stopped for about 2 minutes for part (a), then it will start to rotate for parts (b) and (c). The objects for parts (b) and (c) will be displayed simultaneously.

- (a) A meteor shower will be visible in the sky. Determine the constellation of the radiant and estimate its right ascension and declination coordinates.

1.a (3 pt)

Constellation	right ascension	declination

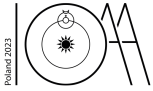
- (b) Identify which of the following variable stars visible in the sky are in low (write 'DIM') or high (write 'BRIGHT') brightness states. The mean magnitude as shown in the atlas and the magnitude range are given for each star.

1.b (8 pt)

Name	atlas mag.	mag. range	DIM / BRIGHT
γ Cas (<i>Cih</i>)	2	1.6 — 3.0	
δ Cep	4	3.5 — 4.4	
μ Cep (<i>Erakis</i>)	4	3.4 — 5.1	
β Per (<i>Algol</i>)	2	2.2 — 3.4	
α Cet (<i>Mira</i>)	3.5	2.0 - 10.1	
χ Cyg	4.5	3.3 - 14.1	
L ₂ Pup	4.5	2.6 — 6	
δ Sco (<i>Dschubba</i>)	2	1.6 — 2.3	

- (c) Identify the constellations whose borders are marked and give their IAU abbreviations.

Planetarium Round



A1-2

English (Official)

1.c (9 pt)